

Arnab Ardhendu Purkayastha, Ph.D.

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Professional Summary

Assistant Professor and engineering researcher working at the intersection of FPGA-based acceleration, high-performance computing, and embedded systems. An engineering educator committed to hands-on, entrepreneurially minded learning. Active in the Kern Entrepreneurial Engineering Network (KEEN) as a participant, presenter, workshop leader, and faculty mentor. Guides undergraduates to peer-reviewed publications and award-winning design projects.

FPGA and reconfigurable computing

High-performance and cloud computing

Hardware acceleration for machine learning at the edge

VLSI design and physical-design optimization

Entrepreneurially minded, project-based engineering education

Degrees

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|------|---|
| 2021 | Ph.D., Electrical & Computer Engineering
University of North Carolina at Charlotte, NC, USA
<i>Dissertation: Empowering Reconfigurable Platforms for Massively Parallel Applications</i> |
| 2013 | M.Tech., Microelectronics & VLSI
National Institute of Technology (NIT) Agartala, India |
| 2011 | B.E., Electrical Engineering
Tripura University, India |

Academic & Professional Appointments

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| Jan 2022 – Present | Assistant Professor, Electrical & Computer Engineering
Western New England University, Springfield, MA <ul style="list-style-type: none">Undergraduate and graduate teaching in computer engineering and first-year engineering; first-year advisor; senior capstone advisor; undergraduate research mentor. |
| Aug 2021 – Dec 2021 | Research Scientist
TeCSAR Labs, UNC Charlotte, NC <ul style="list-style-type: none">High-performance computing applications on cloud and edge FPGAs. |
| 2017 – 2021 | Graduate Research Assistant
UNC Charlotte, NC <ul style="list-style-type: none">Ph.D. research on FPGA parallelism, memory-access optimization, and OpenCL high-level synthesis, including three industry-funded projects. |
| 2017 – 2021 | Graduate Teaching Assistant
UNC Charlotte, NC <ul style="list-style-type: none">Computer Architecture, Heterogeneous Computing, Network Theory I, Programming in C++, and Logic Design for undergraduate and graduate students. |
| Feb 2014 – Jan 2017 | Assistant Professor
MU, Mumbai, India <ul style="list-style-type: none">Taught VLSI Design, IC Technology, and Microcontrollers. |

- PLC programming and HMI interfaces for industrial weighing and control systems.

Peer-Reviewed Work

Journal Articles

1. A. A. Purkayastha, J. Tharwani, S. Aggarwal, "Ant Colony Optimization for CMOS Physical Design: Reducing Layout Area and Improving Aspect Ratio in VLSI Circuits," *Electronics (MDPI)*, vol. 14, no. 24, art. 4825, 2025. doi:10.3390/electronics14244825
2. J. Tharwani, A. A. Purkayastha, "Cost-Performance Evaluation of General Compute Instances: AWS, Azure, GCP, and OCI," *International Journal of Computer Trends and Technology*, vol. 72, no. 11, pp. 248–255, 2024.
3. A. Danesh Pazho, G. Alinezhad Noghre, A. A. Purkayastha, J. Vempati, O. Martin, H. Tabkhi, "A Survey of Graph-Based Deep Learning for Anomaly Detection in Distributed Systems," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, pp. 1–20, 2023. doi:10.1109/TKDE.2023.3282898
Google Scholar Metrics: #1 venue in Databases & Information Systems (h5-index 126).
4. "Current Mirror Using FinFETs," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, 2021.
5. A. A. Purkayastha, S. Rogers, S. A. Shiddibhavi, H. Tabkhi, "LLVM-Based Automation of Memory Decoupling for OpenCL Applications on FPGAs," *Microprocessors and Microsystems (Elsevier)*, vol. 72, art. 102909, 2020. doi:10.1016/j.micpro.2019.102909
Google Scholar Metrics: #6 venue in Computer Hardware Design (h5-index 60).
6. "Improving Performance of VLSI Circuits Using Different Modes of FinFET," *International Journal of Research and Analytical Reviews (IJRAR)*, 2019.

Conference Papers

7. C. Burnham*, A. A. Purkayastha, N. Magotra, "Speculative Backpropagation AI Training Framework for FPGA-Oriented Implementation at the IoT Edge," *69th IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2026.
8. J. Tharwani, S. Aggarwal, A. A. Purkayastha, "Quantifying Virtualization Overhead for CPU-Intensive HPC Workloads in Hypervisors: A Comparative Study of KVM, Proxmox VE, Xen, and Hyper-V," *IEEE SoutheastCon*, pp. 1–6, 2026. doi:10.1109/SoutheastCon63549.2026.11476478
9. A. A. Purkayastha, S. Aggarwal, J. Tharwani, "Heuristic-Driven Resource Optimization for Decoupled Memory Channels on FPGAs," *IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)*, vol. 4, pp. 1–6, 2026. doi:10.1109/IATMSI68868.2026.11465435
10. S. Aggarwal, A. A. Purkayastha, J. Tharwani, "Empowering IoT with Federated Learning: A Cross-Domain Survey from Healthcare to Industry," *IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)*, vol. 4, pp. 1–5, 2026.
11. S. Centeno*, C. Sprague*, A. A. Purkayastha, R. Simar, N. Magotra, "Exploring Parallelism in FPGA-Based Accelerators for Machine Learning Applications," *68th IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, pp. 842–846, 2025. doi:10.1109/MWSCAS53549.2025.11244519
12. A. A. Purkayastha, J. Tharwani, S. Aggarwal, "FPGA or GPU? Analyzing Comparative Research for Application-Specific Guidance," *IEEE SoutheastCon, Concord, NC*, pp. 1258–1263, 2025. doi:10.1109/SoutheastCon56624.2025.10971274
13. J. Tharwani, S. Aggarwal, A. A. Purkayastha, "Evaluating HPC-Style CPU Performance and Cost in Virtualized Cloud Infrastructures," *IEEE SoutheastCon, Concord, NC*, pp. 1239–1244, 2025. doi:10.1109/SoutheastCon56624.2025.10971705
14. J. Ekong, A. A. Purkayastha, G. Ekong, "Work in Progress: Impact of Collaborative Learning Strategies on Anxiety Reduction in Introductory Programming Courses," *ASEE Annual Conference & Exposition*, 2024.
15. S. Centeno*, A. A. Purkayastha, "Beyond Trial and Error: Automating Delay Generation in Assembly with Algorithmically Calculated For-Loops," *Global Interdisciplinary Green Cities Conference*, 2024.
16. S. Aggarwal, A. A. Purkayastha, "SecureTrack: A Contact Tracing IoT Platform for Monitoring Infectious Diseases," *Global Interdisciplinary Green Cities Conference*, 2023.

17. A. A. Purkayastha, H. Tabkhi, "Design Study on Impact of Memory Access Parallelism for Cloud FPGAs," *34th IEEE International System-on-Chip Conference (SOCC)*, pp. 254–259, 2021. doi:[10.1109/SOCC52499.2021.9739477](https://doi.org/10.1109/SOCC52499.2021.9739477)
18. J. Thiagarajan, A. A. Purkayastha, A. Patil, H. Tabkhi, "Exploring the Scalability of OpenCL Coarse Grained Parallelism on Cloud FPGAs," *33rd IEEE International System-on-Chip Conference (SOCC)*, pp. 213–218, 2020. doi:[10.1109/SOCC49529.2020.9524765](https://doi.org/10.1109/SOCC49529.2020.9524765)
19. A. A. Purkayastha, S. Raghavendran, J. Thiagarajan, H. Tabkhi, "Exploring the Efficiency of OpenCL Pipe for Hiding Memory Latency on Cloud FPGAs," *IEEE High Performance Extreme Computing Conference (HPEC)*, pp. 1–7, 2019. doi:[10.1109/HPEC.2019.8916236](https://doi.org/10.1109/HPEC.2019.8916236)
20. A. A. Purkayastha, S. A. Shiddhibhavi, H. Tabkhi, "Taxonomy of Spatial Parallelism on FPGAs for Massively Parallel Applications," *31st IEEE International System-on-Chip Conference (SOCC)*, pp. 55–60, 2018. doi:[10.1109/SOCC.2018.8618501](https://doi.org/10.1109/SOCC.2018.8618501)

Teaching Scholarship

21. A. A. Purkayastha, "Remote-Controlled Deep Work Sign: AI-Enhanced Product Design," *KEEN Engineering Unleashed*, Card #6116, 2026.

Book

22. "VLSI Design," *Tech-Neo Publications*, ISBN 978–93–90376–04–9, 2020.

* Undergraduate student mentee.

Research Projects & Outcomes

Industry funded IEEE TKDE 2023 Anomaly Detection in Complex Distributed Systems Graph Isomorphism Networks to find and mitigate malicious software processes and gadget chains when hardware/software components of a large system are updated.	IEEE SOCC 2021 Memory Access Parallelism for Cloud FPGAs Formalized Double Data Rate and burst-transfer optimization to cut global-memory latency on Xilinx cloud FPGAs.
Industry funded IEEE SOCC 2020 Automating Parallelism on Cloud FPGAs A generic template and Python-based tool for OpenCL design-space exploration on FPGAs.	Elsevier MICPRO 2020 LLVM-Based Memory Decoupling on FPGAs An LLVM automation tool that decouples memory access from computation on Intel Stratix V FPGAs: over 2× speedup and over 40% energy reduction on Rodinia benchmarks.
Industry funded IEEE HPEC 2019 OpenCL Pipes on Cloud FPGAs Evaluated memory-access decoupling with OpenCL pipes on Xilinx cloud FPGA benchmark applications.	IEEE SOCC 2018 Taxonomy of Parallelism on FPGAs A taxonomy formalizing how OpenCL temporal and spatial parallelism map onto FPGAs.

Grants

KEEN Faculty Development Workshop Grant (Co-PI) "Transform Teaching with EM and Technology – Make Learning Fun Again!"	\$12,500	2025
Davis Educational Foundation Grant (Co-PI) "Effect of Pedagogical Interventions on Student Motivations and Attitudes in Quantitative Courses"	\$11,500	2024
Rice University & Octavo Systems FPGA hardware for the Ghost-in-Machine research collaboration	\$2,500	2025
WNE Summer Research Grants (2×) ENGR 103 Arduino curriculum redesign	\$5,000	2023, 2024

WNE Honors Course Creation Grant LaTeX for technical writing with Overleaf	\$1,200
Total	\$32,700

Proposals Submitted

- **NSF Engineering Research Initiation (ERI), CCSS program:** “LLVM-Guided ML for Memory-Aware Hardware Selection in Heterogeneous Systems: Automatic Application-to-Hardware Mapping across CPU, GPU, and FPGA” (submitted; not funded; resubmission planned)

Courses Taught

Western New England University: Undergraduate

CPE 310	Microprocessors I CISC-to-RISC microprocessor systems; revamped with new material in 2023–24.	Fall 2022 – Fall 2025
CPE 360	Microprocessors II Redesigned in 2025 around the ATmega328PB: firmware-centric labs on interrupts, peripherals, and device drivers; AMD industry guest lecture.	Spring 2022 – Spring 2025
ENGR 103	Introduction to Engineering Arduino curriculum redesign (2023–24) and a Raspberry Pi Pico 2 W robotics platform for ResQTech-themed design challenges and Botbash.	Fall 2022 – Fall 2025
ENGR 105	Computer Programming for Engineers MATLAB; redesigned with Write-Pair-Share, pair programming, and vertical non-permanent surfaces (Davis project).	Spring 2022 – Spring 2025
ENGR 110	Data Acquisition and Processing New preparation: LabVIEW and Raspberry Pi Pico data acquisition with an original AI-enhanced product-design module.	Spring 2026
CPE 491/591	Special Topics: VLSI Design Full physical-design flow (partitioning to routing, SRAM) with DSCH/Microwind labs and an algorithm-implementation project.	Spring 2024, Spring 2025

Western New England University: Graduate

CPE 691	Special Topics: VLSI Design New course created; later cross-listed as CPE 491/591.	Spring 2023
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Engineering Education & Pedagogy

Kern Entrepreneurial Engineering Network (KEEN), 2022 – Present

2022	ICE 1.0 STEM Faculty Teaching Workshop (6 days), Denver, CO — participant
2023	KEEN STEM Faculty Training Conference (3 days), Atlanta, GA — participant
2024	ICE 2.0 Workshop (6 days), Phoenix, AZ — participant and presenter
2025	KEEN National Conference, Austin, TX — participant
2025	Co-organized and led a 3-day KEEN faculty-development workshop at WNE for new College of Engineering faculty; led the Arduino experiential-learning session and mentored participants as they implemented course innovations
2025	KEEN Leader Meeting, Milwaukee, WI — participant

Pedagogical Development & Teaching Innovation

- Davis Educational Foundation project (IRB-approved): implemented and studied active-learning interventions in ENGR 105.
- Co-facilitated the Center for Teaching & Learning workshop “Balancing Content Delivery and Student Engagement in Challenging Courses” (Nov 2024).

- Created the WNE Breadboard Simulator (apwne.github.io/WNE-FY-Breadboard-Simulator) and the WNE-Bot WiFi remote-control platform for first-year engineering.
- Teaching development: CTL New Faculty Orientation and Difficult Conversations (2022); JEDI NITE (2023); Mental Health First Aid (2023); Social & Emotional Learning, CITI IRB certification, and CliftonStrengths (2024).

Student Research & Design Mentoring

WNE Jet Engine ECU

[Lucian Dupont-LeBlanc](#) · Senior capstone · 2025–26

First Prize, Best Senior Design Project in ECE (2026)

- Controls and engine-control-unit architecture for a cross-departmental capstone with Mechanical Engineering and Pratt & Whitney.

Speculative Backpropagation for FPGA-Based Edge AI

[Cameron Burnham](#) · Senior capstone · 2025–26

IEEE MWSCAS 2026

- Reusable benchmarking platform for speculative backpropagation on FPGA targets.

Parallelism in FPGA-Based ML Accelerators

[Sed Centeno & Christopher Sprague](#) · Senior capstone · 2024–25

IEEE MWSCAS 2025

- Rice University Ghost-in-Machine collaboration: ~15% faster neural-network training on an Octavo Systems FPGA board.

Car Theft Alarm System

[Logan Paranto](#) · Senior capstone · 2023–24

- Battery-powered intrusion detector using a pressure sensor, buzzer alarm, and real-time GPS location on an ESP32 microcontroller.

IoT-Controlled Desk for Immersive Tabletop Gaming

[Corbin Veitch & Tim Fortune](#) · Senior capstone · 2023–24

- Home Assistant-based desk controlling lights and sound over MQTT from a single embedded processor, with custom Python libraries and UIs.

Botbash First-Year Robotics Competition

[ENGR 103 student teams](#)

First place (2025); third place (2024); second place (2023)

- Advised first-year teams designing robots for ResQTech disaster-response challenges on the Pico 2 W platform.

Invited Talks

Invited research presentation: FPGA-based memory-channel optimization and resource-efficient computing architectures

[ABV-Indian Institute of Information Technology and Management \(ABV-IIITM\), Gwalior, India \(virtual\), 2026](#)

“Accelerators & AI: Optimizing Energy for a Sustainable Future”

[AICTE Training and Learning \(ATAL\) Academy Faculty Development Program on Energy, Sustainability and Climate Change, India \(online\), Jan 16–22, 2025](#)

“Electronics Systems: Devices, IC and Applications”

[AICTE Short-Term Training Program, Department of ECE, NIT Arunachal Pradesh, India, Mar 14–18, 2022](#)

Professional Service

Conference Leadership

- Technical Track Chair, Global Interdisciplinary Green Cities Conference: 2023 (Augsburg, Germany), 2024 (Springfield, MA), 2025 (Pittsburgh, PA), 2026 (Western New England University)
- Technical Track Chair, 67th IEEE International Midwest Symposium on Circuits and Systems (MWSCAS), 2024
- Technical Program Committee, Security, Surveillance and Artificial Intelligence Conference (ICSSAI), Techno India University, 2023 and 2025

Journal Reviewer

- IEEE Transactions on Neural Networks and Learning Systems (2)
- IEEE Transactions on Circuits and Systems
- Microprocessors and Microsystems (Elsevier)
- The Journal of Supercomputing (Springer, 3)
- Scientific Reports (Springer Nature)
- Artificial Intelligence Review (Springer)
- Journal of Intelligent Information Systems (Springer)
- Circuits, Systems, and Signal Processing (Springer)
- Silicon (Springer)
- Journal of Low Power Electronics (ASP, 2)

Conference & Other Reviewing

- IEEE MWSCAS 2025 (3 papers) and 2026
- IEEE ISIE 2025
- Int'l Conference on VLSI Design & Embedded Systems 2024 (4 papers)
- Global Interdisciplinary Green Cities Conference 2024
- External Ph.D. dissertation reviewer, Vignan University, India

University & Community

- Engineers Week (E-Week): hands-on robot demonstrations and student-engagement activities (2024, 2025)
- College of Engineering Outreach Committee: Open House, Accepted Students Day, ECE Major Showcase, Homecoming
- Judge: Western Massachusetts Science & Engineering Fair (Region 1); [FIRST Robotics](#) Tech Challenge, WNE Qualifier; WNE Engineering Expo

Awards & Honors

- **Richard Newton Young Fellow Award** — Design Automation Conference (DAC), San Francisco, CA, 2018
- **KFF Scholarship** — KEEN ICE 2.0 Workshop, 2024
- **GASP Scholarship** — UNC Charlotte (Ph.D.)
- **MHRD Scholarship** — Government of India (M.Tech.)
- **NEC Fellowship** — Government of India (B.E.)
- **First Prize, Microprox** — Inter-collegiate technical festival, 2011

Technical Skills

AI & machine learning	FPGA acceleration of neural-network training; graph neural networks; federated learning; generative-AI-assisted engineering workflows (code development, technical writing, course design) and teaching students to use AI critically
Programming	C++, OpenCL, CUDA, MATLAB, Python
Hardware description	Verilog, VHDL
Hardware platforms	Intel/Altera Stratix V and Arria 10 FPGAs; Xilinx Alveo and Virtex-5 FPGAs; AMD FirePro W7100 GPU; Raspberry Pi Pico 2 W; ATmega328PB; Arduino
Cloud	AWS EC2 F1 (Xilinx VU9P), NSF Chameleon Cloud FPGAs
Tools	Xilinx Vivado, Vitis and ISE; Intel HLS; Cadence Virtuoso; ModelSim; OrCAD PSpice; DSCH; Microwind; Tanner; LabVIEW; Keil μ Vision
Operating systems	Linux (Ubuntu, CentOS), Windows